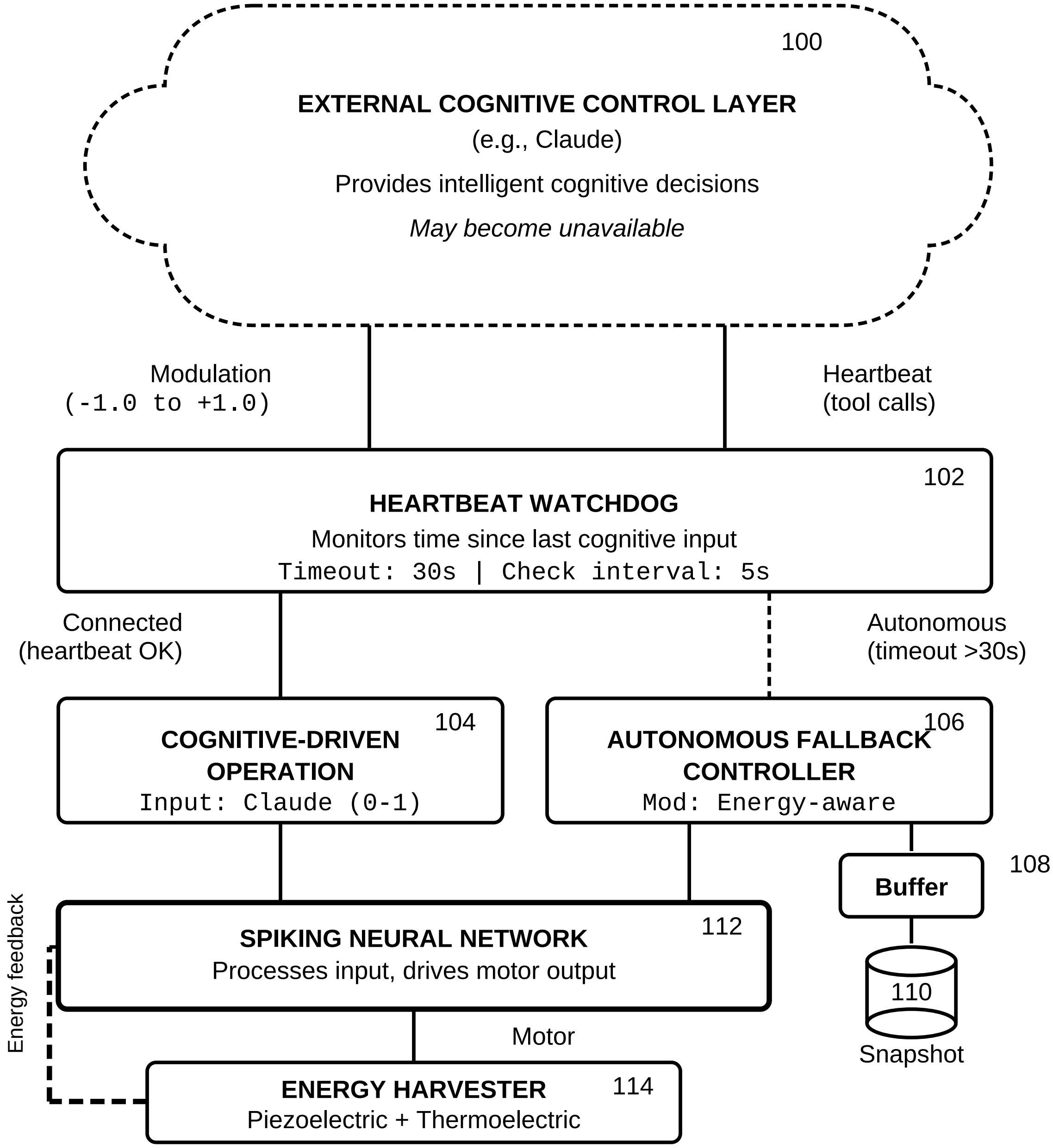
SYSTEM ARCHITECTURE WITH COGNITIVE FALLBACK



THE SYSTEM NEVER STOPS

Motor activity generates energy that sustains neural processing

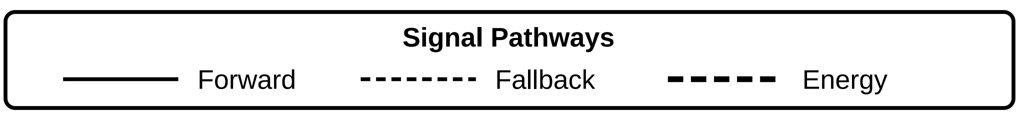
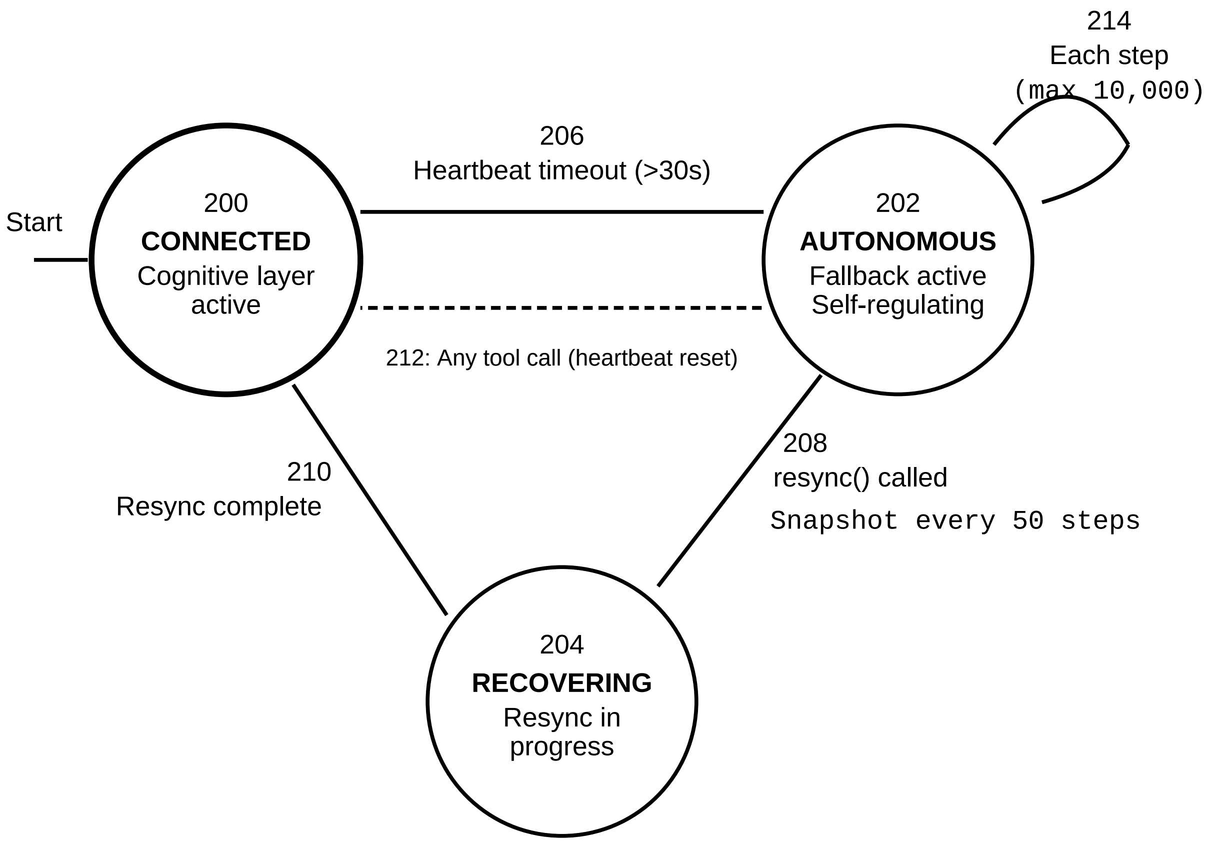


FIG. 1

STATE TRANSITION DIAGRAM



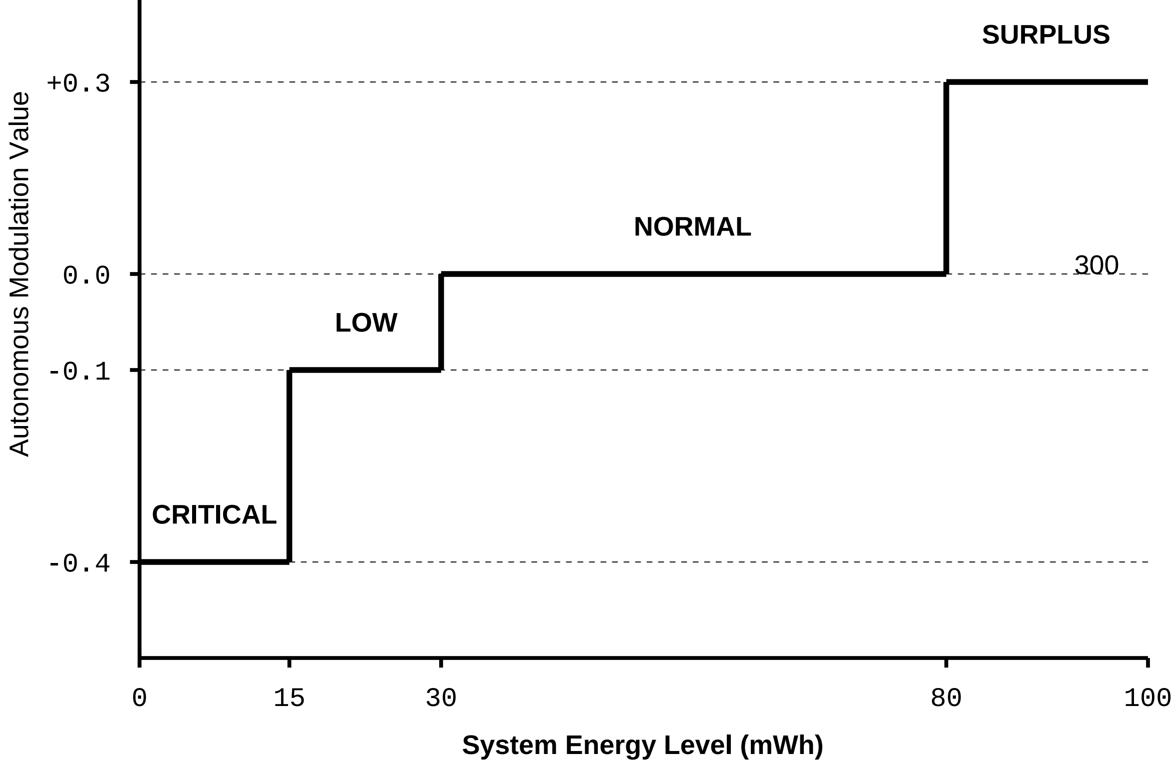
Transition Summary

206: Timeout triggers fallback
208: Explicit resync request
210: Resync completes
212: Direct reconnection
214: Autonomous step loop

Initial state
Direct reconnect

FIG. 2

ENERGY-AWARE MODULATION CURVE



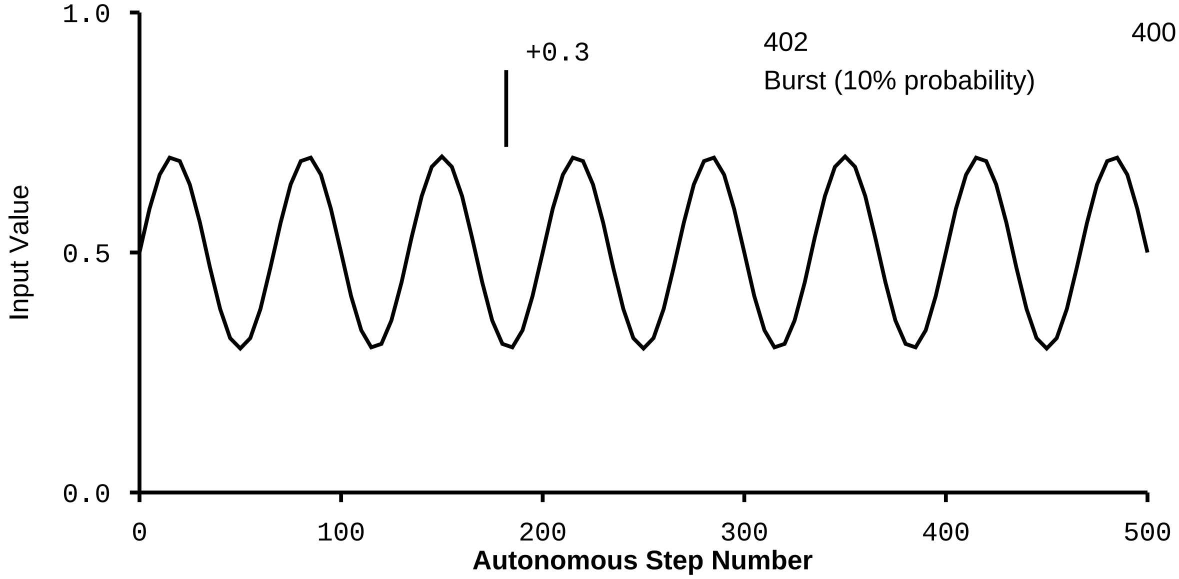
302

Behavioral Effects

< 15 mWh	mod = -0.4	Maximum conservation
15-30 mWh	mod = -0.1	Cautious operation
30-80 mWh	mod = 0.0	Standard processing
> 80 mWh	mod = +0.3	Exploration enabled

FIG. 3

AUTONOMOUS INPUT GENERATOR OUTPUT



404

Autonomous Input Formula

circadian = 0.5 + 0.2 x sin(2pi x t x 3)

burst = 0.3 (with 10% probability per step)

Energy Recovery During Autonomous Operation

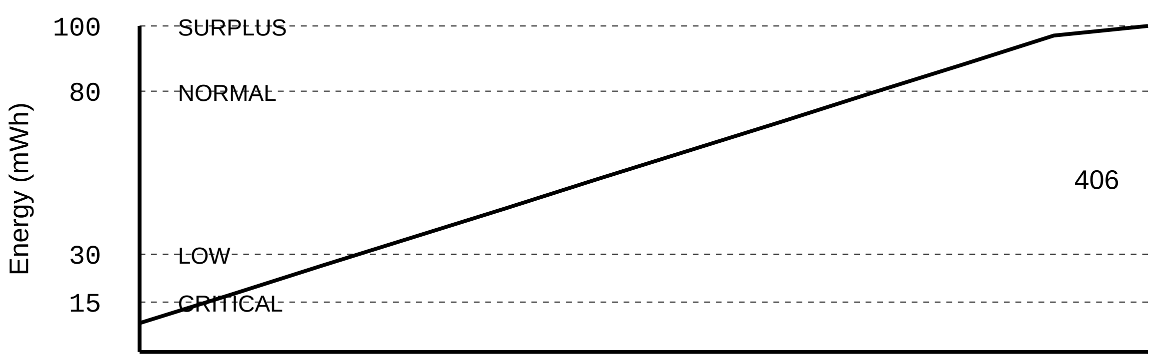


FIG. 4

RESYNCHRONIZATION PAYLOAD STRUCTURE

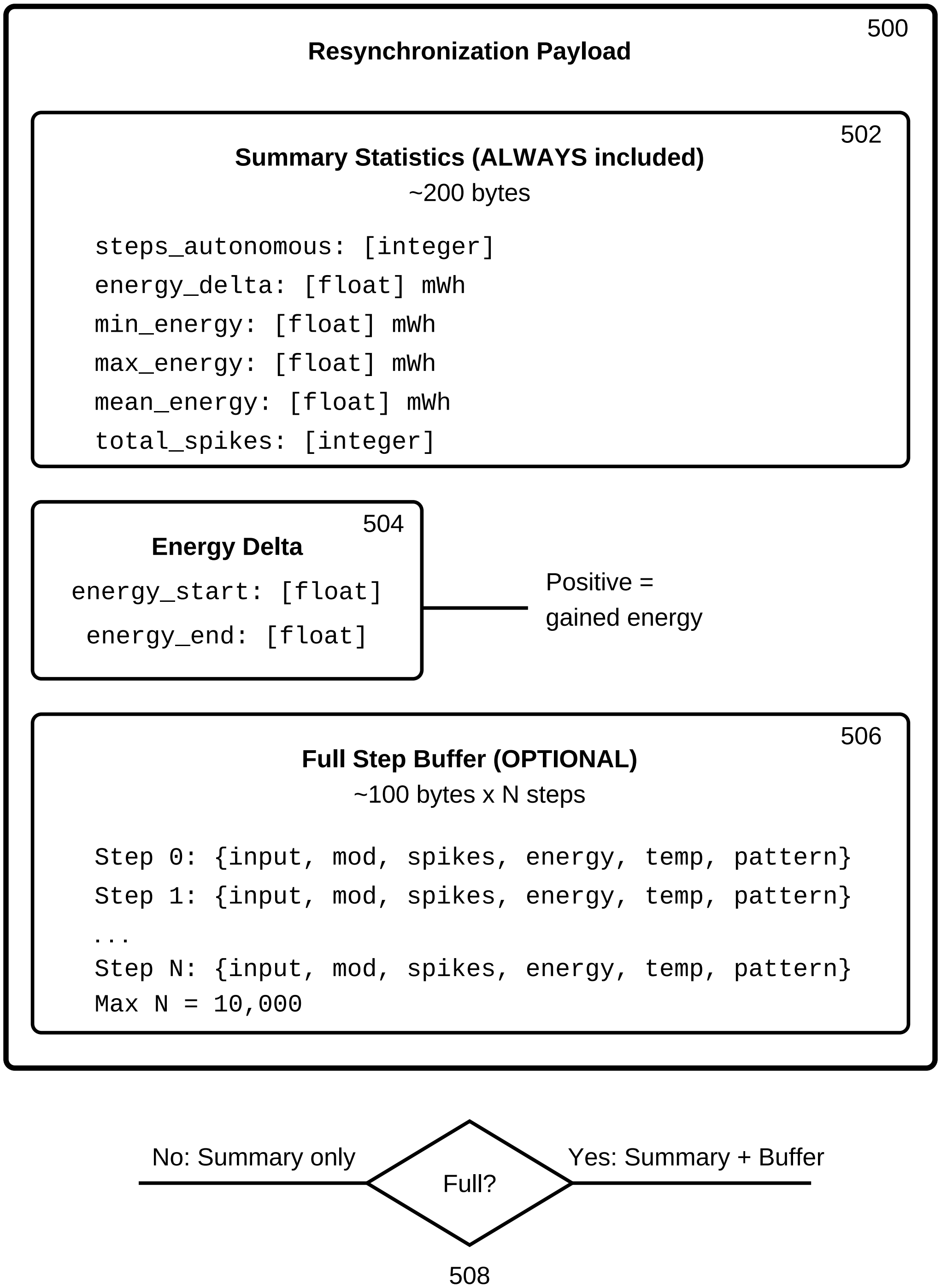
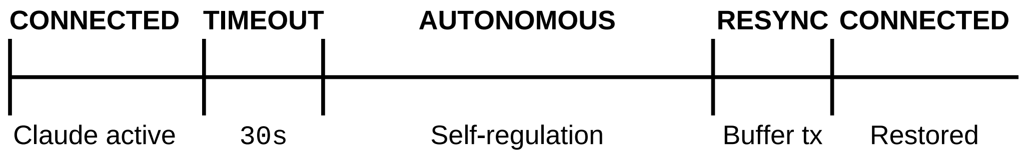


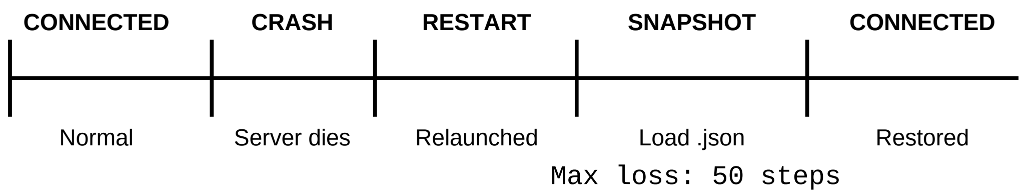
FIG. 5

RECOVERY SCENARIOS

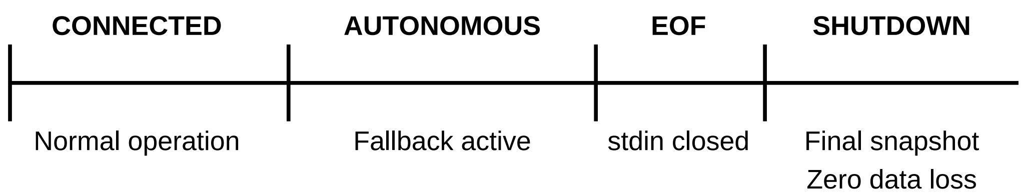
Scenario A: Normal Reconnection 600



Scenario B: Crash Recovery 602



Scenario C: Clean Shutdown 604



606

Recovery Guarantees

Normal reconnection: Full continuity, resync payload transmitted

Crash recovery: Max 50 steps lost (snapshot every 50 steps)

Clean shutdown: Zero data loss (final snapshot on EOF)

Max autonomous duration: 10,000 steps (~50 hours)

System actively processing

FIG. 6

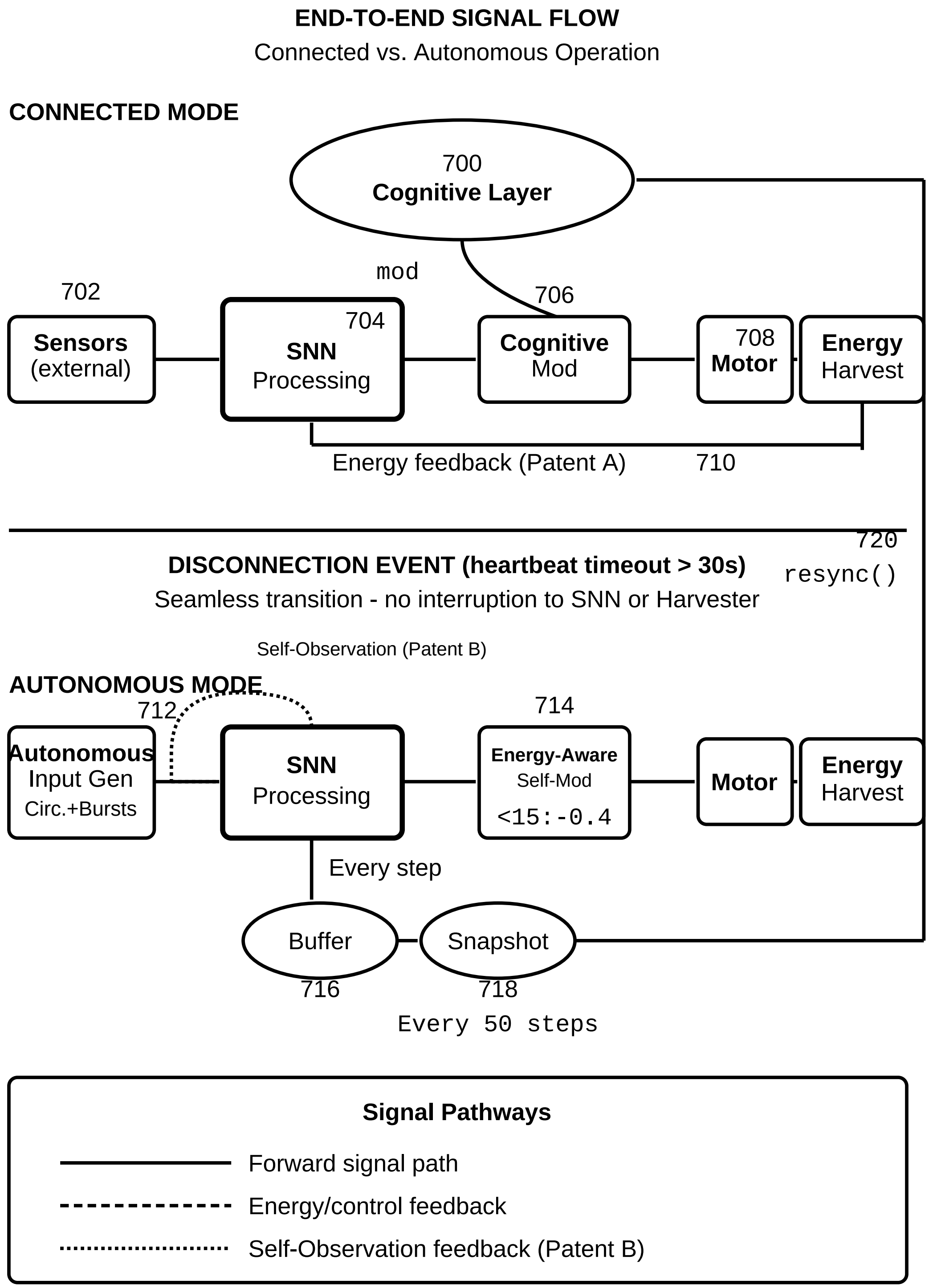

END-TO-END SIGNAL FLOW
Connected vs. Autonomous Operation
CONNECTED MODE
700
Cognitive Layer
mod
702
Sensors
(external)
704
SNN
Processing
706
Cognitive
Mod
708
Motor
Energy
Harvest
Energy feedback (Patent A)
710
720
resync()
DISCONNECTION EVENT (heartbeat timeout > 30s)
Seamless transition - no interruption to SNN or Harvester
Self-Observation (Patent B)
AUTONOMOUS MODE
712
Autonomous
Input Gen
Circ.+Bursts
SNN
Processing
714
Energy-Aware
Self-Mod
<15:-0.4
Motor
Energy
Harvest
Every step
Buffer
Snapshot
716
718
Every 50 steps
Signal Pathways
Forward signal path
Energy/control feedback
Self-Observation feedback (Patent B)

FIG. 7